MICRO POWER

JUMPER-PAD-2-NO
DIODE-SCHOTTKY_SMBRA140
D5
PTCTHERMISTOR1206
F1
U2 LM1117
IN OUT
ADJ
C11 10uF
C10 10uF
C9 0.1uF
R16 240
R31 1KOHM-0603-1/10W-1%
D6 RED
R15 715
R18 390
JUMPER-PAD-3-2OF3_NC_BY_PASTE
J2
5V<-
->3.3V
VCC
GND
RAW

INPUT VOLTAGE 7-16V
Varies depending on testbed

V1 (3.3 or 5)

JUMPER-PAD-3-2OF3_NC_BY_PASTE
J3
U3 LM1117
C6 10uF
C5 10uF
C4 0.1uF
C3 100uF
R4 240
R1 715
R7 390
35/V1_5V
33/V1_3.3V
MOSFET-PCHANNELNDS6020P_20V_24A_LOW_VGS_TH
Q3
R10 10K
R5 1KOHM-0603-1/10W-1%
C17 0.1uF
D1 RED
R11 100
Q2 MMBT2222A
48/PWR_CTRL_1

V2 (3.3, 3.7, 4.2, 5)

JUMPER-PAD-3-2OF3_NC_BY_PASTE
J4
U4 LM1117
C15 10uF
C14 10uF
C13 0.1uF
C8 100uF
R19 240
R21 715
R8 390
R36 470
R35 470
R34 100
37/V2_3.7V
36/V2_3.3V
38/V2_4.2V
39/V2_5V
MOSFET-PCHANNELNDS6020P_20V_24A_LOW_VGS_TH
Q5
R12 10K
R20 1KOHM-0603-1/10W-1%
C18 0.1uF
D4 RED
R14 100
Q4 MMBT2222A
49/PWR_CTRL_2

MICRO

R2 10K
SWITCH-MOMENTARY-2SMD-REDUNDANT
S1
C2 0.1uF
DTR
RESET
Y1 16MHZ
U1
RESET
XTAL2
XTAL1
AREF
AVCC
AGND
VCC
GND
C22 0.1uF
C21 0.1uF
C19 0.1uF
C1 0.1uF

ATMEGA2560V

R6 1KOHM-0603-1/10W-1%
D3 RED

1.8-5V
3.3V
EN_I2C
TX1
RX1
SCL1
SDA1
U5
EN
VREF1
VREF2
SCL1
SCL2
SDA1
SDA2
GND
PCA9306
1-3.6V

JP3
AVR_SPI_PRG_6PTH
MISO
SCK
RESET
MOSI

2x30 RA connector to testbeds

FLYING_JALAPENO_V3FEMALE
RAW
V1
V2
A0
A1
A2
A3
A4
A5
A6
A7
A8
A9
A10
A11
A12
A13/RAW_READ
A14/PT_READ_V1
A15/PT_READ_V2
0(RX0)
1(TX0)
2(INT0)
3(INT1)
4
5
6
7
8
9
10
11
12
13
14(TX3)
15(RX3)
16(TX2)
17(RX2)
18(TX1_INT5)
19(RX1_INT4)
20(SDA/INT3)
21(SCL/INT2)
22
23
24
25
26
27
28
29
30
33/V1_3.3V
34/RAW_PASS_LED
35/V1_5V
36/V2_3.3V
37/V2_3.7V
38/V2_4.2V
39/V2_5V
40/FAIL
41/PASS
42/PT_FAIL
43/PT_PASS
44/PT_CTRL
48/PWR_CTRL1
49/PWR_CTRL2
50(MISO)
51(MOSI)
52(SCK)
53(SS)
R25 1KOHM-0603-1/10W-1%
R24 1KOHM-0603-1/10W-1%
R23 1KOHM-0603-1/10W-1%
R22 1KOHM-0603-1/10W-1%
CAP_PAD1
CAP_PAD2
R26 22M
R27 22M
45/CAPSENSE1
47/CAPRETURN1
46/CAPSENSE2
31/CAPRETURN2

SERIAL PROGRAMMING

FTDI_BASIC_DEVICE_SIDERA_SMT
JP2
GND
CTS
VCC
RXI
TXO
DTR
FTDI_VCC
Q1 MOSFET-NCHANNELBSS138
Q6 MOSFET-NCHANNELBSS138
R17 10K
R33 10K
R32 10K
R37 10K

3.3V Source for PCA9306

U6
IN OUT
EN
GND BP
3.3V MIC5205
C12 10UF-16VDC-20%(0603)(TANT)"
C7 0.1UF
C20 10NF-0603-50V-10%

Max Voltage Input: 16VDC
Max Current Output: 150mA

PRETESTS

R13 10K
R3 10K
R9 100K
R30 10K
R28 10K
R29 100K

sparkfun
open hardware

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TITLE: ${PROJECTNAME}
Design by: Pete Lewis
REV: v11
Sheet: /
File: ${FILENAME}.sch
Date: ${CURRENT_DATE}
Sheet: ${#}/${##}
Title:
Size: User
Date:
Rev:
KiCad E.D.A. eeschema 7.0.6
Id: 1/1